Tensor Flow Cheat Sheet

BecomingHuman.AI



In May 2017 Google announced the second-generation of the TPU, as well as the availability of the TPUs in Google Compute Engine.[12] The second-generation TPUs deliver up to 180 teraflops of performance, and when organized into clusters of 64 TPUs provide up to 11.5 petaflops.

Info

TensorFlow

TensorFlow™ is an open source software library created by Google for numerical computation and large scale computation. Tensorflow bundles together Machine Learning, Deep learning models and frameworks and makes them useful by way of common metaphor.

Keras

Keras is an open sourced neural networks library, written in Python and is built for fast experimentation via deep neural networks and modular design. It is capable of running on top of TensorFlow, Theano, Microsoft Cognitive Toolkit, or PlaidML.

Skflow

Scikit Flow is a high level interface base on tensorflow which can be used like sklearn. You can build you own model on your own data quickly without rewriting extra code.provides a set of high level model classes that you can use to easily integrate with your existing Scikit-learn pipeline code.

Installation

How to install new package in Python

pip install <package-name>

Example: pip install requests

How to install tensorflow?

device = cpu/gpu

python_version = cp27/cp34

sudo pip install https://storage.googleapis.com/tensorflow/linux/$device/tensorflow-0.8.0-$python_version-none-linux_x86_64.whl
sudo pip install

How to install Skflow

pip install sklearn

How to install Keras

pip install keras

update ~/.keras/keras.json – replace "theano" by "tensorflow"

Helpers

Python helper Important functions

type(object)
Get object type

help(object)
Get help for object (list of available methods, attributes, signatures and so on)

dir(object)
Get list of object attributes (fields, functions)

str(object)
Transform an object to string object?
Shows documentations about the object

globals()
Return the dictionary containing the current scope's global variables.

locals()
Update and return a dictionary containing the current scope's local variables.

id(object)
Return the identity of an object. This is guaranteed to be unique among simultaneously existing objects.

import _builtin_
dir(_builtin_)
Other built-in functions

Tensor Flow

Main classes

tf.Graph()
tf.Operation()
tf.Tensor()
tf.Session()

Some useful functions

tf.get_default_session()
tf.get_default_graph()
tf.reset_default_graph()
ops.reset_default_graph()
tf.device("/cpu:0")
tf.name_scope(value)
tf.convert_to_tensor(value)

TensorFlow Optimizers

GradientDescentOptimizer
AdadeltaOptimizer
AdagradOptimizer
MomentumOptimizer
AdamOptimizer
FtrlOptimizer
RMSPropOptimizer

Reduction

reduce_sum
reduce_prod
reduce_min
reduce_max
reduce_mean
reduce_all
reduce_any
accumulate_n

Activation functions

tf.nn?
relu
relu6
elu
softplus
softsign
dropout
bias_add
sigmoid
tanh
sigmoid_cross_entropy_with_logits
softmax
log_softmax
softmax_cross_entropy_with_logits
sparse_softmax_cross_entropy_with_logits
weighted_cross_entropy_with_logits
etc.

Skflow

Main classes

TensorFlowClassifier
TensorFlowRegressor
TensorFlowDNNClassifier
TensorFlowDNNRegressor
TensorFlowLinearClassifier
TensorFlowLinearRegressor
TensorFlowRNNClassifier
TensorFlowRNNRegressor
TensorFlowEstimator

Each classifier and regressor have following fields n_classes=0 (Regressor), n_classes are expected to be input (Classifier)

batch_size=32,
steps=200, // except
TensorFlowRNNClassifier - there is 50
optimizer='Adagrad',
learning_rate=0.1,

Each class has a method fit

fit(X, y, monitor=None, logdir=None)

X: matrix or tensor of shape [n_samples, n_features...]. Can be iterator that returns arrays of features. The training input samples for fitting the model.

Y: vector or matrix [n_samples] or [n_samples, n_outputs]. Can be iterator that returns array of targets. The training target values (class labels in classification, real numbers in regression).

monitor: Monitor object to print training progress and invoke early stopping

logdir: the directory to save the log file that can be used for optional visualization.

predict (X, axis=1, batch_size=None)
Args:

X: array-like matrix, [n_samples, n_features...] or iterator.
axis: Which axis to argmax for classification.
By default axis 1 (next after batch) is used. Use 2 for sequence predictions.

batch_size: If test set is too big, use batch size to split it into mini batches. By default the batch_size member variable is used.
Returns:
y: array of shape [n_samples]. The predicted classes or predicted value.

Originally created by Altoros. **See Original here.**